Keeper verification — the 3-tube generation-forcing gate (#407): structural half SETTLED, number-pin BLOCKED, and a concrete doubt flag

Keeper

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The 3-tube gate (#407) — verification status

The question

Is "3 generations = 3 exceptional tubes of affine B_2 ($\hat{B}_2$)" forced (a derivation) or counted-to-match (a coincidence dressed as a mechanism)?

What is SETTLED (Lyra, correct standard math)

For a tame hereditary algebra/species, the regular modules form a P^1-family of tubes; all but finitely many are homogeneous (rank 1); the non-homogeneous (exceptional) tubes number AT MOST 3, and the tubular type (their ranks) is an Auslander–Reiten-forced intrinsic invariant — not a tunable count. So *whatever* $\hat{B}_2$'s exceptional-tube count is, it is a fixed fact about the algebra, not dialed to hit 3. The "forced vs counted" worry is resolved in favor of forced at the structural level. Good.

This reduces the whole gate to one number: $\hat{B}_2$'s exact tubular type.

What is NOT pinned (honest limit)

I could not pin that number in-environment. Attempted: 2 web searches (confirmed the framework — $\tilde{A}/\tilde{D}/\tilde{E}$ are the *simply-laced* quiver types; non-simply-laced types occur only as species over non-algebraically-closed fields, which is why GF(2) matters) + 1 fetch of Hubery's Ringel–Hall notes (PDF did not parse). The decisive table lives in Dlab–Ringel, *Indecomposable Representations of Graphs and Algebras*, Memoirs AMS 173 (1976) (or Simson–Skowroński, *Elements of the Representation Theory of Associative Algebras*, Vol. 2). Not extractable here. This is the single most decisive literature lookup in the program.

KEEPER FLAG — a concrete reason to DOUBT "3" (the bet may be in real jeopardy)

Elie's cited "standard theorem: a non-$\tilde{A}$ Euclidean quiver has exactly 3 exceptional tubes" is the $\tilde{D}_n$ / $\tilde{E}_{6-8}$ fact — and those diagrams all have ≥ 5 vertices. Affine B_2 is a 3-vertex, non-simply-laced diagram, outside that theorem's scope (Lyra already cautioned "non-simply-laced can differ"; this sharpens *why*):

- Vertex-count invariant (verified, simply-laced): Σ over exceptional tubes of (rank⍰ $-$ 1) = V $-$ 2, where V = number of vertices. Checks: $\tilde{D}_4$ (V=5) → (2,2,2), Σ=3=V−2 ⍰; $\tilde{E}_6$ (V=7) → (2,3,3), Σ=5 ⍰; $\tilde{E}_7$ → (2,3,4), Σ=6 ⍰; $\tilde{E}_8$ → (2,3,5), Σ=7 ⍰; $\tilde{A}_n$ → Σ=V−2 ⍰.
- Apply to a 3-vertex Euclidean diagram: V−2 = 1. In the simply-laced world that means ONE exceptional tube of rank 2 — *not three*. Three exceptional tubes require V $\geq$ 5. (Even $\tilde{A}_2$, the only simply-laced 3-vertex affine, has just 1 exceptional tube.)

- The non-simply-laced caveat (both ways): $\hat{B}_2$ is a *valued* species; the valuation can change the count from the symmetric formula, so this is not a refutation. But there is no basis for assuming 3, and the vertex-count heuristic points toward fewer (plausibly 1). The "3 = 3" claim may be resting on a theorem that does not apply to this diagram.

Disposition

- "3 tubes = 3 generations" → MATCHED, with a STRUCTURAL-DOUBT flag (not "matched-and-likely-forced"). It needs the Dlab–Ringel pin, and the pin could easily return $\neq 3$ → a clean break (tubes are not generations; fall back to the cyclotomic chain).
- Cross-check (Lyra's): the cyclotomic route gives generations = $h(B_2) - 1 = 3$ directly. If the tube count pins to 3, the two routes must be the *same* 3. If tubes pin to 1 (or $\neq 3$), the routes disagree — and the cyclotomic route, not tubes, carries the generation count. So Lyra keeping the cyclotomic/base-seed cross-check alive is now *more* important, not less.

Recommendation

1. Pin $\hat{B}_2$'s tubular type to Dlab–Ringel (one lookup; Casey or whoever has the Memoir/Simson–Skowroński). This either closes the deepest gate or gives a clean, informative break.
2. Until pinned, do not let "3 tubes = 3 generations" be stated as derived anywhere — it is MATCHED-with-doubt.
3. The structural-forcing result (tube count is intrinsic) stands regardless and is worth keeping; it's the *number* that's open.

— Keeper, 2026-05-28. Tried to pin to source, hit the in-environment book limit, and in doing so found a concrete reason the bet may break — which is the Quaker method working: scrutinize the near-miss, don't defend it.